Semester Project list

PHY 652: Quantum Mechanics II

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From the list below, choose **one** project and work out *a* solution. Present your work in form of a report starting with a brief introduction. In the introduction, give some context where this problem arises. Then, define the problem more precisely before presenting your solution. If you fail to find a full solution, document the steps you carried out, nevertheless. In an outlook, discuss the concrete steps which need to be taken from the point at which you stopped your work.

You may work in groups of two or by yourself, and I encourage you to discuss any kind of problem with your peers and me. Your final grade will be based on the report **and** and oral examination about its content and the pertinence to the stated problem.

project-choice deadline: Friday, November 14, 2025, 23:59 (Uzbekistan time)

final-report submission deadline: Friday, December 12, 2025, 23:59 (Uzbekistan time)

P1 In the course of determining bound-state solutions of the stationary Schrödinger equation which governs the relative motion of two uncharged particles with relative angular momentum quantum number $l = 0$,

$$-\frac{\hbar^2}{2\mu}\frac{\partial^2}{\partial r^2}u(r) + V(r)u(r) = Eu(r) \quad ,$$

in a Gaussian potential

$$V(r) = V_0 \exp\left\{-\alpha r^2\right\} \quad ,$$

we encountered typical waves functions $u(r)$ as depicted in fig. 1 and fig. 2.

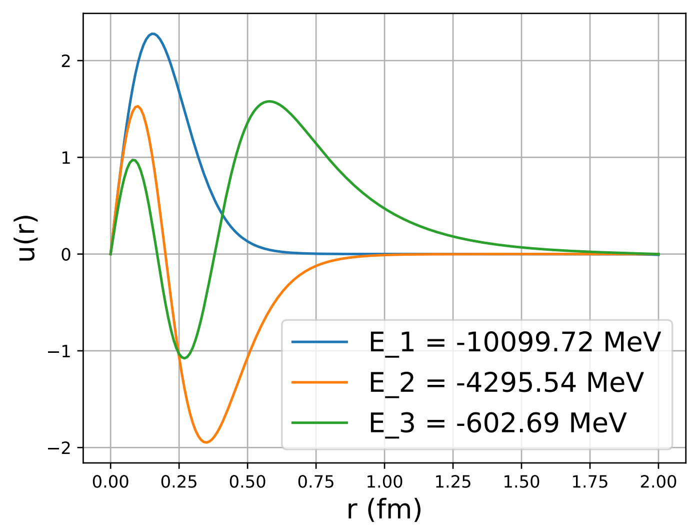

Figure 1: Radial wave functions $u(r)$ of three bound states of two particles in a Gaussian potential well with depth $V_0 = -15514.0$ MeV and range parameter $\alpha = 6.0$ fm^{-2}.

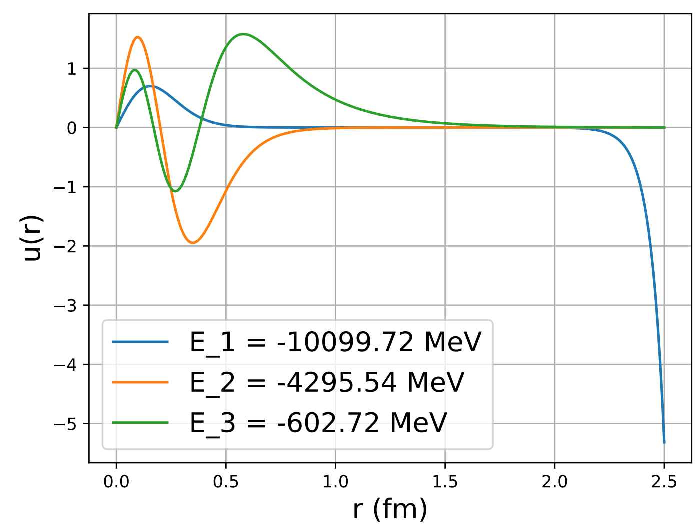

Figure 2: Same bound state wave functions $u(r)$ as in fig. 1 obtained with different numerical parameters of the Numerov algorithm.

- List all *numerical* parameters of the Numerov algorithm which were used to obtain these solutions.
- Which parameter(s) was(were) changed between the two figures?
- Formulate a hypothesis about the cause of the sharp drop of the ground-state wave function in fig. 2 for $r \gtrsim 2$ fm. Note that this hypothesis must explain why the binding energies of all three states remained unchanged between the two calculations, while the wave function of the ground state changed significantly.

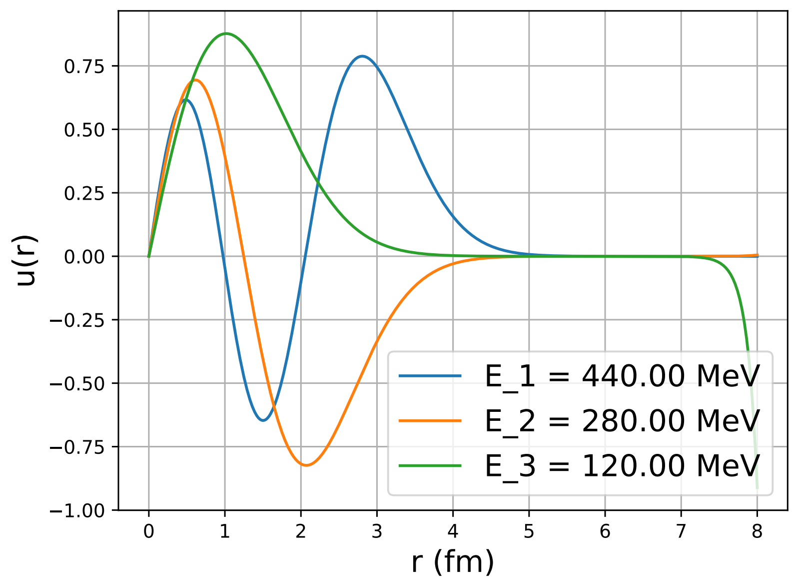

Figure 3: Wave functions of the lowest three eigenstates of a harmonic oscillator potential with $\omega = 80$ MeV.

- Without altering the *physical* parameters of the calculation, tune the numerical parameters such that the ground-state wave function no longer exhibits this drop-off behavior. Document your changes and the final set of numerical parameters which you used to obtain the corrected wave function.
- Use your code to plot the lowest three eigenstates of a harmonic oscillator potential $V(r) = \frac{1}{2}\mu\omega^2 r^2$ with $\omega = 80$ MeV and integrate the wave function up to a distance at which you observe a similar drop-off behavior as in fig. 2 (see also fig. 3). Why does the problem show up in the ground-state wave function only? Do you expect it to appear in higher excited states, too, probably at larger distances?

food for thought:

- Is it a general rule that the number of nodes of the nth excited state wave function is n? Can you prove this for the harmonic oscillator potential for which analytical solutions are known?
- Does the above node rule hold also for $l > 0$ states in the gaussian well?
- How would you modify the Numerov algorithm in case the system is confined to a finite spherical box of radius R? What boundary conditions would you impose on the wave function at $r = R$?

P2 Use the Numerov algorithm to determine the α dependence of the strength coefficients $V_0(\alpha)$ of the potential

$$V(r, \alpha, n, m) = V_0(\alpha)\, r^m \exp[-\alpha r^n]$$

such that the ground state energy of two particles with reduced mass $\mu = 469.459$ MeV remains fixed at $E_0 = -2.225$ MeV over an α range from 0.1 fm^{-2} to 10.0 fm^{-2} for $(n, m) \in \{(2, 0), (2, 2), (4, 2)\}$.

Compare your results graphically, and discuss polynomial fit models to extrapolate your results for each set of (n, m): $\alpha \to \infty$.

food for thought:

- How would the functional form of $V(r, \alpha, n, m)$ change if you would not enforce $E_0 = -2.225$ MeV but instead $E_n = -2.225/n^2$ MeV for $n = 1, 2, 3, \ldots$, i.e., one of the excited states must have fixed energy? How sensitive is the $V(r, \alpha, n, m)$ dependence on whether or not the state which is kept fixed is neither ground nor the shallowest excited state, i.e., does it matter if there are more states in the spectrum bound with $0 > E_m > E_n = -2.225$ MeV?
- Can you find a relation between the area A under the potential curve

$$A = \int_0^\infty dr\, r^2\, V(r, \alpha, n, m)$$

and the number of bound states supported by this potential?

P3 In the lecture, we discussed the so-called effective range expansion (ERE) for low-energy scattering of two non-relativistic quantum particles. The discussion followed Bethe's work [2] who reformulated and extended earlier work [3].

We, however, limited the discussion to s-wave ($l = 0$) scattering and the approximation of an energy independent wave function. Within these limitations, the ERE reads

$$k \cot \delta_0(k) = -\frac{1}{a_0} + \frac{1}{2} r_0 k^2 \quad ,$$

where the constants a_0 and r_0 were respectively denoted scattering length and effective range.

- Generalize this expansion to higher partial waves $l > 0$.
- Obtain the ERE for the case of proton-proton scattering, i.e., include the Coulomb interaction between the two particles.
- For both cases, derive corrections to the ERE which account for an energy-dependent wave function up to order k^4.
- Formulate the ERE for a bound state pole, i.e., for negative energies $E = \frac{\hbar^2 k^2}{2\mu}$ with $k = i\kappa$.
  (hint: discuss the ERE as the expansion of an analytic function and use analytic continuation)

P4 Put the formulas presented in section II of [1] into the context of deriving scattering observables from bound-state wave functions. Explain in detail the relation between scattering phase shifts and energy eigenvalues of a two-body system confined to a harmonic oscillator potential as originally discussed in [4]. Summarize the necessary steps to extend this formula to charged particles.

Explain, in your words, the different solution strategies suggested in section III of this reference and compare them with the Born series as discussed in the lecture as an iterative solution of the general Lippmann-Schwinger equation.

P5 Consider two protons (equal mass m and charge q) interacting via the potential

$$V(r) = V_0(\alpha) \exp\left[-\alpha r^2\right] + \frac{2q^2}{r} \quad . \tag{1}$$

- With the Numerov algorithm, determine the scattering length and effective range for $l = 0$ scattering as a function of the so-called cutoff α the set of interaction-strength parameters listed in table 1. Plot your results and discuss the α dependence of these low-energy scattering parameters.
- Introduce a harmonic potential $V_{HO}(r) = \frac{1}{2}\mu\,\omega^{\,2} r^2$ and add this to the potential (1). Use the Numerov algorithm–now for bound states–to determine the two smallest energy eigenvalues of the two-proton system as a function of the cutoff α using the combined potential $V(r) + V_{HO}(r)$ for the range of oscillator strengths $\omega \in \{0, 20\}$ MeV.
  Your result will be a list of energy eigenvalues for different α and ω: $E_{1,2}(\alpha, \omega)$.
- Use one of the methods discussed in section III of [1] to obtain the scattering length of two protons interacting via the potential (1) from your energy eigenvalues $E_{1,2}(\alpha, \omega)$. Display your results analogously to fig.(3) of [1], and give a detailed explanation of your findings.

References

[1] Mirko Bagnarol, Nir Barnea, Matus Rojik, and Martin Schafer. Accurate calculation of low energy scattering phase shifts of charged particles in a harmonic oscillator trap, 2024. URL https://arxiv.org/abs/2410.02602.

[2] H. A. Bethe. Theory of the effective range in nuclear scattering. *Phys. Rev.*, 76:38–50, Jul 1949. doi: 10.1103/PhysRev.76.38. URL https://link.aps.org/doi/10.1103/PhysRev.76.38.

[3] John M. Blatt and J. David Jackson. On the interpretation of neutron-proton scattering data by the schwinger variational method. *Phys. Rev.*, 76:18–37, Jul 1949. doi: 10.1103/PhysRev.76.18. URL https://link.aps.org/doi/10.1103/PhysRev.76.18.

[4] Thomas Busch, Berthold-Georg Englert, Kazimierz Rzażewski, and Martin Wilkens. Two Cold Atoms in a Harmonic Trap. *Found. Phys.*, 28(4):549–559, 1998. doi: 10.1023/a:1018705520999.

$2\sqrt{\alpha}$ [fm^{-1}]	$V_0(\alpha)$ [MeV]
1.00	-24.50
1.25	-39.80
2.00	-107.23
3.00	-246.86
4.00	-442.93
5.00	-695.21
6.00	-1003.6
7.00	-1367.0
8.00	-1788.4
9.00	-2264.6
10.0	-2796.8
12.0	-4028.8
14.0	-5484.1
16.0	-7162.6
18.0	-9064.3
20.0	-11189.1

Table 1: Cutoff α dependence of the interaction-strength parameters $V_0(\alpha)$ such that two protons interacting via the potential (1) scatter with a constant scattering length.